

The Language of Medicine

13th edition

Davi-Allen Chabner

Chapter 9

Male Reproductive System

Chapter Goals (Slide 1 of 2)

- Name, locate, and describe the functions of the organs of the male reproductive system.
- Define abnormal conditions and infections that affect the male reproductive system.
- Differentiate among several types of sexually transmitted infections.

Chapter Goals (Slide 2 of 2)

- Define combining forms used to describe the structures of this system.
- Describe various laboratory tests and clinical procedures that are pertinent to the disorders of the male reproductive system and recognize related abbreviations.
- Apply your new knowledge to understanding medical terms in their contexts, such as medical reports and records.

Chapter 9

Lesson 9.1

Introduction (Slide 1 of 4)

- The male sex cell is called a **spermatozoon**, or sperm cell.
- The sperm cell carries its nuclear hereditary material, or chromosomes, in its head region.
- The tail consists of a **flagellum**, which makes the sperm cell mobile.

Introduction (Slide 2 of 4)

- Only one spermatozoon out of 300 million sperm cells released during ejaculation can fertilize a single female ovum.
- If more than one egg is present in the fallopian tubes during ejaculation, multiple fertilizations are possible.

Introduction (Slide 3 of 4)

- Twins resulting from the fertilization of separate ova by separate sperm cells are called fraternal twins.
- Twins resulting from the fertilization of a single egg by a single sperm are called identical twins.

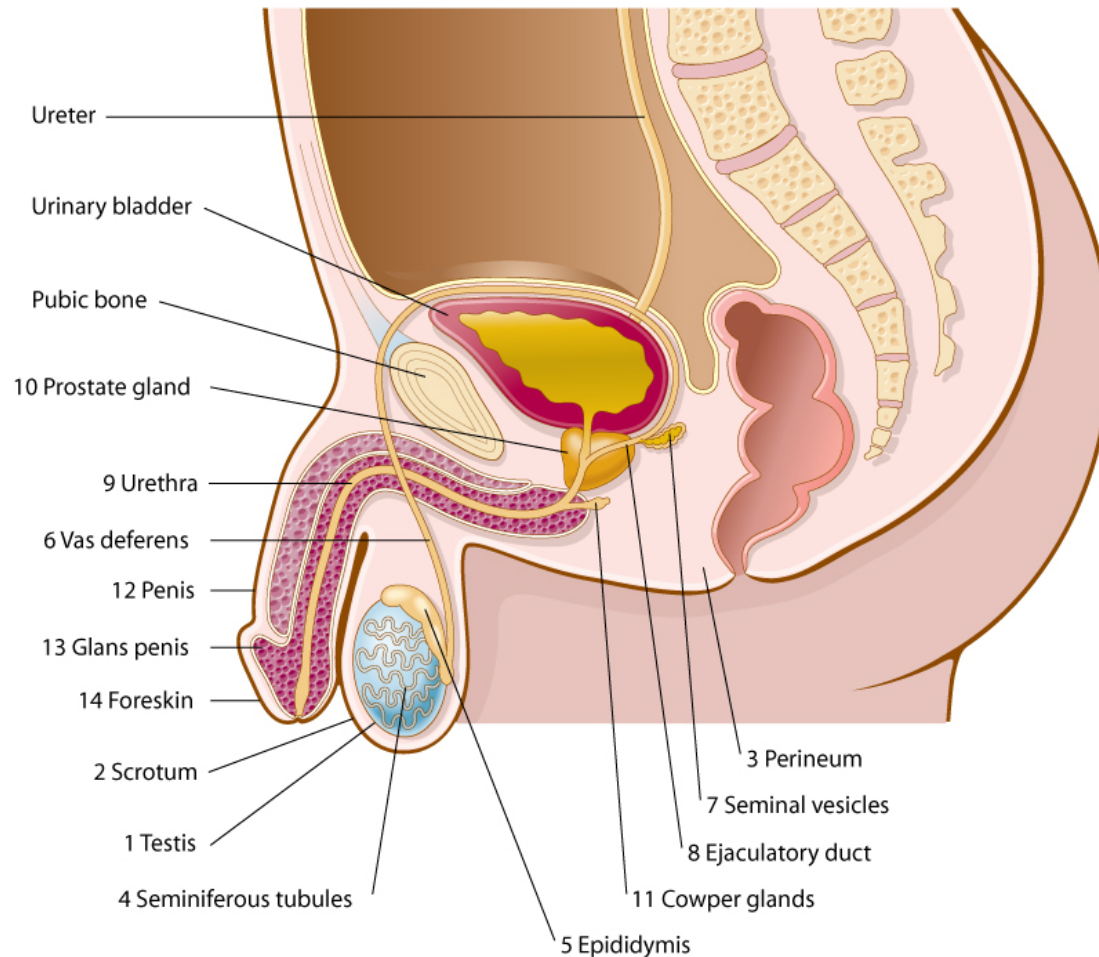
Introduction (Slide 4 of 4)

- The organs of the male reproductive system are designed to produce and release billions of sperm cells throughout the man's lifetime.
- The hormone called **testosterone** is responsible for the bodily characteristics of the male (such as beard, pubic hair, and deeper voice).

Anatomy of the Male Reproductive System (slide 1 of 4)

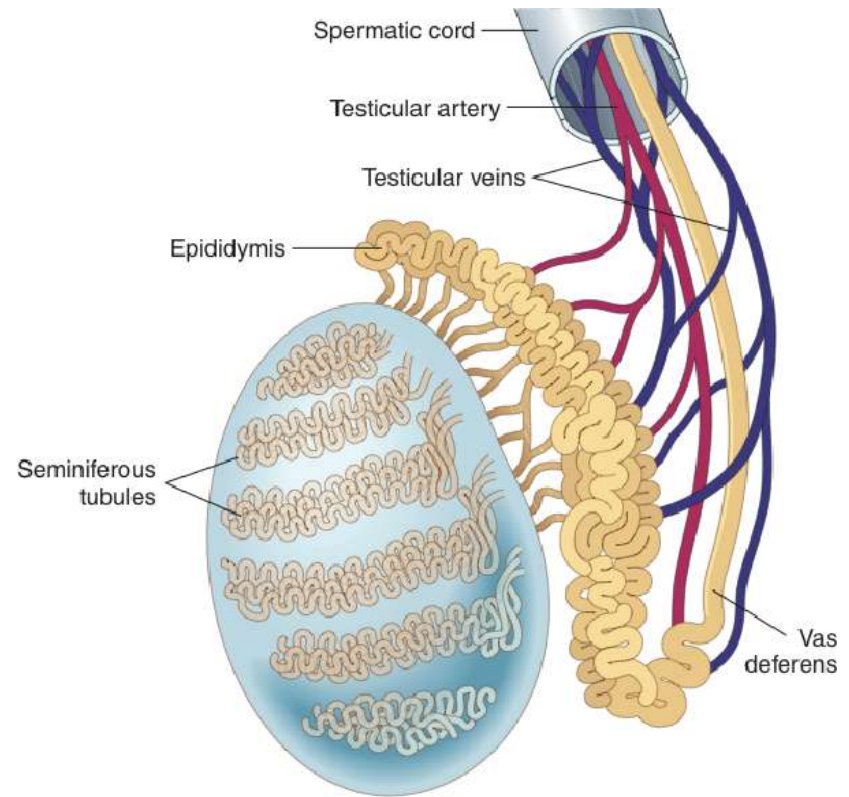
- Testis
- Scrotum
- Perineum
- Seminiferous tubules
- Epididymis
- Vas deferens
- Seminal vesicles
- Ejaculatory duct
- Urethra
- Prostate gland
- Bulbourethral glands
- Penis
- Glans penis
- Prepuce (foreskin)

Anatomy of the Male Reproductive System (slide 2 of 4)



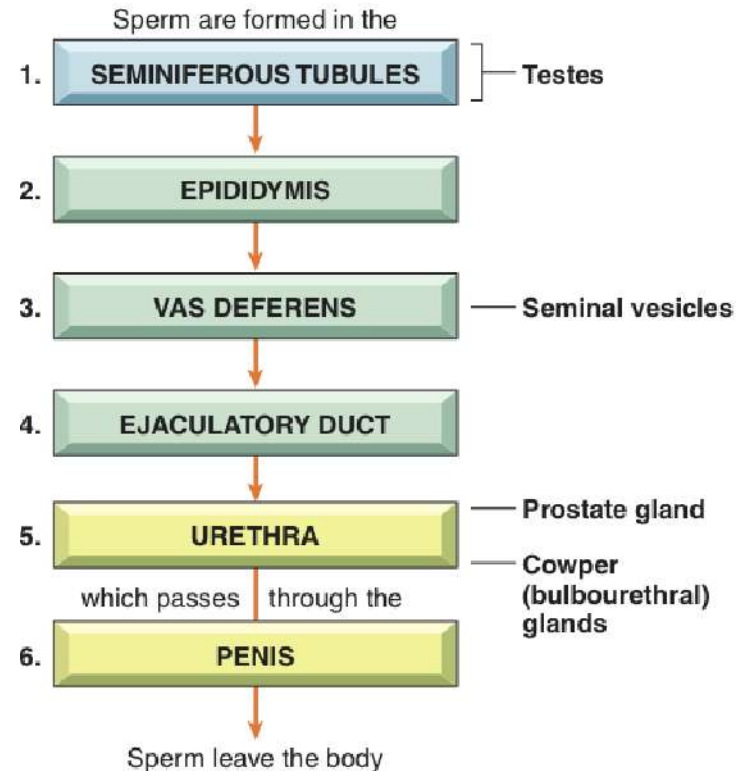
Anatomy of the Male Reproductive System (slide 3 of 4)

- Internal structure of the testis and the epididymis



Anatomy of the Male Reproductive System (slide 4 of 4)

- The passage of sperm from the seminiferous tubules to the outside of the body



Vocabulary (Slide 1 of 16)

Term
bulbourethral glands
ejaculation
ejaculatory duct

Vocabulary (Slide 2 of 16)

Term	Meaning/Definition
bulbourethral glands	Pair of exocrine glands near male urethra; Cowper glands
ejaculation	Ejection of sperm and fluid from male urethra
ejaculatory duct	Tube through which semen enters the male urethra

Vocabulary (Slide 3 of 16)

Term
epididymis (<i>pl.</i> : epididymides)
erectile dysfunction
flagellum

Vocabulary (Slide 4 of 16)

Term	Meaning/Definition
epididymis (<i>pl.</i> : epididymides)	One of a pair of long, tightly coiled tubes above each testis that stores and carries sperm to the vas deferens
erectile dysfunction	Inability of an adult male to achieve an erection; impotence
flagellum	Hair-like projection on a sperm cell that makes it motile

Vocabulary (Slide 5 of 16)

Term
fraternal twins
glans penis
identical twins

Vocabulary (Slide 6 of 16)

Term	Meaning/Definition
fraternal twins	Two infants resulting from fertilization of two separate ova by two separate sperm cells
glans penis	Sensitive tip of the penis; comparable to the clitoris in the female
identical twins	Two infants resulting from division of one fertilized egg into two distinct embryos; conjoined (“Siamese”) twins are incompletely separated identical twins

Vocabulary (Slide 7 of 16)

Term
interstitial cells of the testes
parenchymal tissue
penis

Vocabulary (Slide 8 of 16)

Term	Meaning/Definition
interstitial cells of the testes	Specialized cells adjacent to the seminiferous tubules in the testes; produce testosterone; also called Leydig cells
parenchymal tissue	Essential distinctive cells of an organ
penis	Male external organ of reproduction

Vocabulary (Slide 9 of 16)

Term
perineum
prepuce
prostate gland
scrotum

Vocabulary (Slide 10 of 16)

Term	Meaning/Definition
perineum	External region between the anus and scrotum in the male
prepuce	Foreskin; fold of skin covering tip of penis
prostate gland	Exocrine gland at base of male urinary bladder; secretes fluid part of semen into urethra during ejaculation
scrotum	External sac that contains the testes

Vocabulary (Slide 11 of 16)

Term
semen
seminal vesicles
seminiferous tubules

Vocabulary (Slide 12 of 16)

Term	Meaning/Definition
semen	Spermatozoa and seminal fluid discharged from the urethra during ejaculation
seminal vesicles	Paired sac-like exocrine glands that secrete fluid into the vas deferens
seminiferous tubules	Narrow, coiled tubules that produce sperm in the testes

Vocabulary (Slide 13 of 16)

Term
spermatozoon (<i>pl. spermatozoa</i>)
sterilization
stromal tissue

Vocabulary (Slide 14 of 16)

Term	Meaning/Definition
spermatozoon (<i>pl.</i> spermatozoa)	Sperm cell
sterilization	Procedure that removes a person's ability to produce or release reproductive cells through removal of testicles, vasectomy, or oophorectomy
stromal tissue	Supportive, connective tissue or an organ; stroma

Vocabulary (Slide 15 of 16)

Term
testis (<i>plural: testes</i>)
testosterone
vas deferens

Vocabulary (Slide 16 of 16)

Term	Meaning/Definition
testis (<i>plural: testes</i>)	Male gonad (testicle) that produces spermatozoa and testosterone
testosterone	Hormone secreted by interstitial tissue of the testes; responsible for male sex characteristics
vas deferens	Narrow tube carrying sperm from the epididymis toward the urethra; ductus deferens

QUICK QUIZ (Slide 1 of 4)

1. Which term refers to the skin covering the tip of the penis?

- A. Stroma
- B. Vas deferens
- C. Prepuce
- D. Seminal vesicle

Terminology – Combining Forms

(Slide 1 of 2)

Combining Form	Meaning
andr/o	male
balan/o	glans penis
cry/o	cold
crypt/o	hidden
epididym/o	epididymis
gon/o	seed
hydr/o	water, fluid
orch/o, orchi/o, orchid/o	testis, testicle

Terminology – Combining Forms

(Slide 2 of 2)

Combining Form	Meaning
pen/o	penis
prostat/o	prostate gland
semin/i	semen, seed
sperm/o	spermatozoa, semen
terat/o	monster
test/o	testis, testicle
varic/o	varicose veins
vas/o	vessel duct
zo/o	animal life

Terminology – Suffixes (Slide 1 of 1)

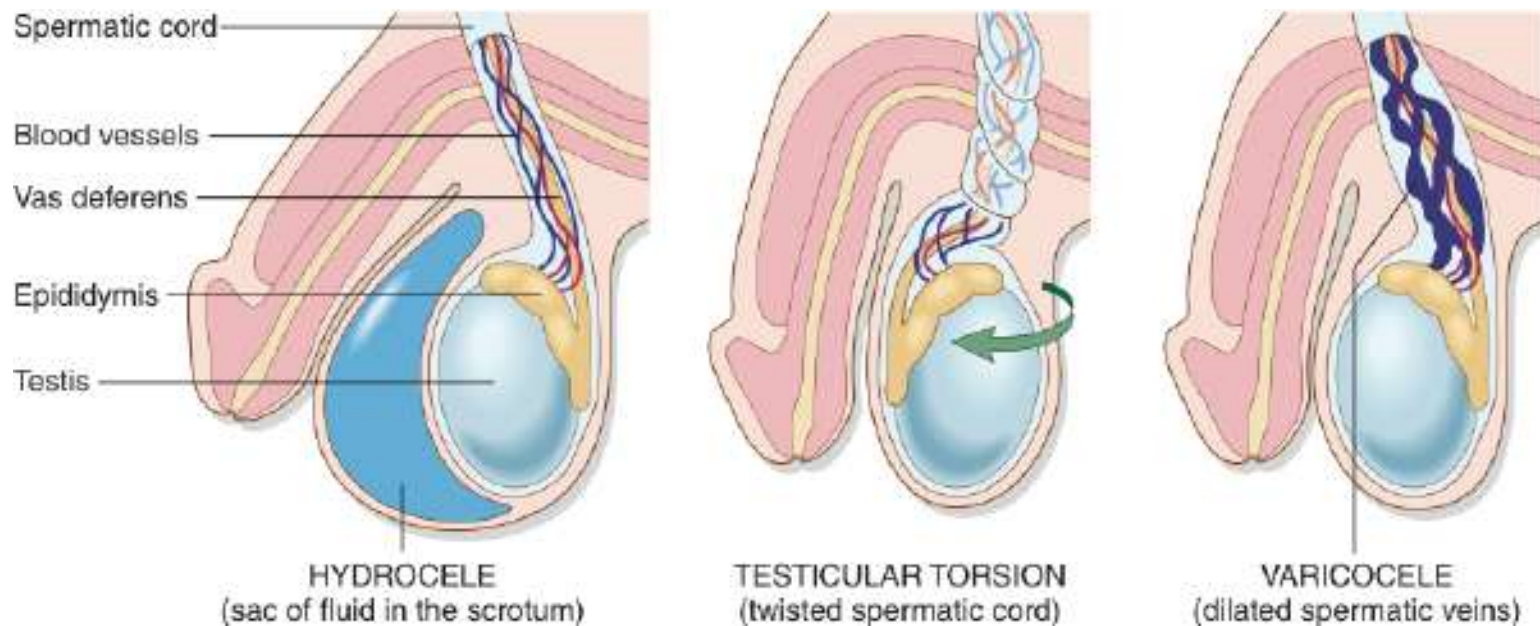
Suffix	Meaning
-genesis	formation
-one	hormone
-pexy	fixation, put in place
-stomy	new opening

Chapter 9

Lesson 9.2

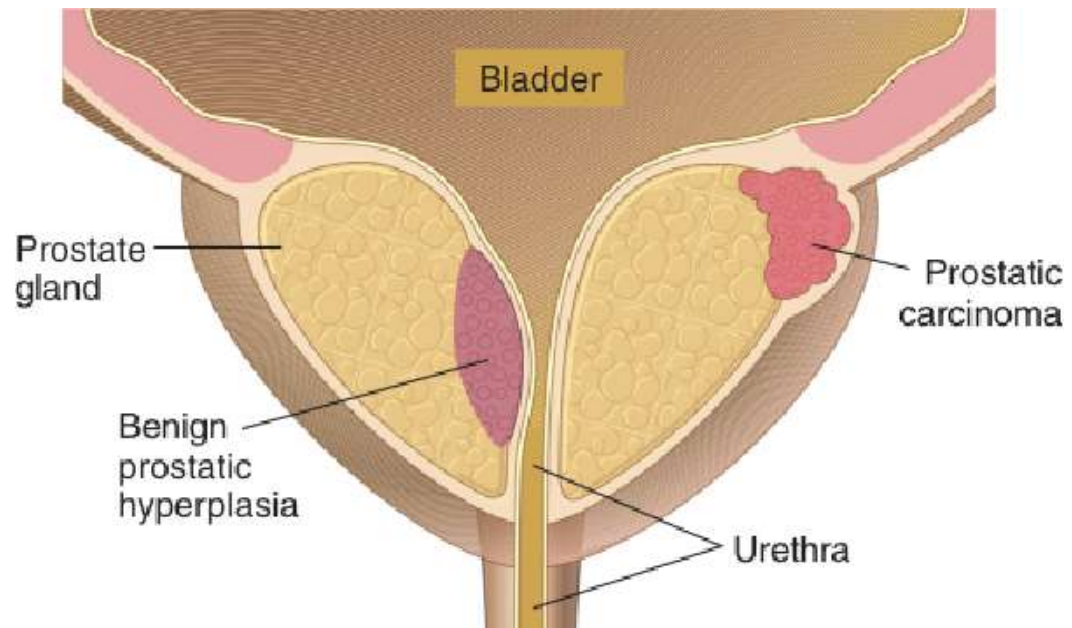
Pathologic Conditions (Slide 1 of 3)

- Hydrocele, Testicular Torsion, and Varicocele



Pathologic Conditions (Slide 2 of 3)

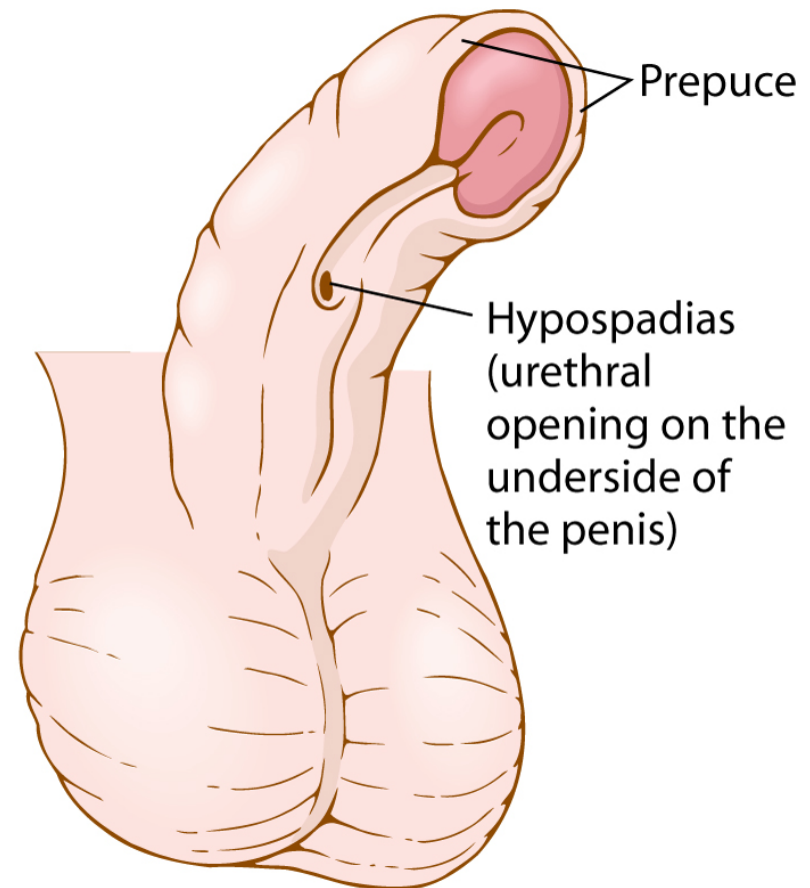
- Prostate Gland with Carcinoma and BPH



Pathologic Conditions (Slide 3 of 3)

Hypospadias

- Surgical repair involves excising a portion of the prepuce, wrapping it around a catheter, suturing it to the distal part of the urethra, bringing it to the end of the penis



Sexually Transmitted Infections

STIs

- Chlamydial infection
- Gonorrhea
- Herpes genitalis
- Human papilloma virus (HPV)
- Syphilis

Laboratory Tests and Clinical Procedures

- **Laboratory tests**

- PSA test

- Semen analysis

QUICK QUIZ (Slide 2 of 4)

2. What does the PSA test for?

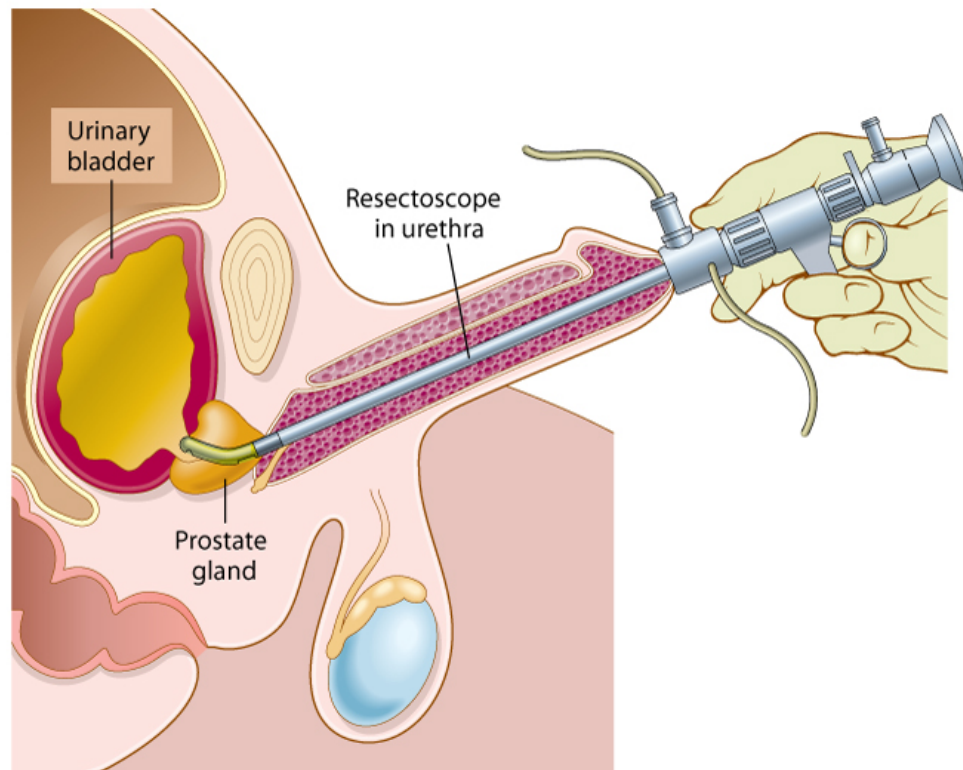
- A. Levels of infection
- B. Sperm mobility
- C. Sperm motility
- D. Levels of an antigen

Laboratory Tests and Clinical Procedures

(Slide 1 of 2)

Clinical procedures

- Transurethral resection of the prostate—TURP

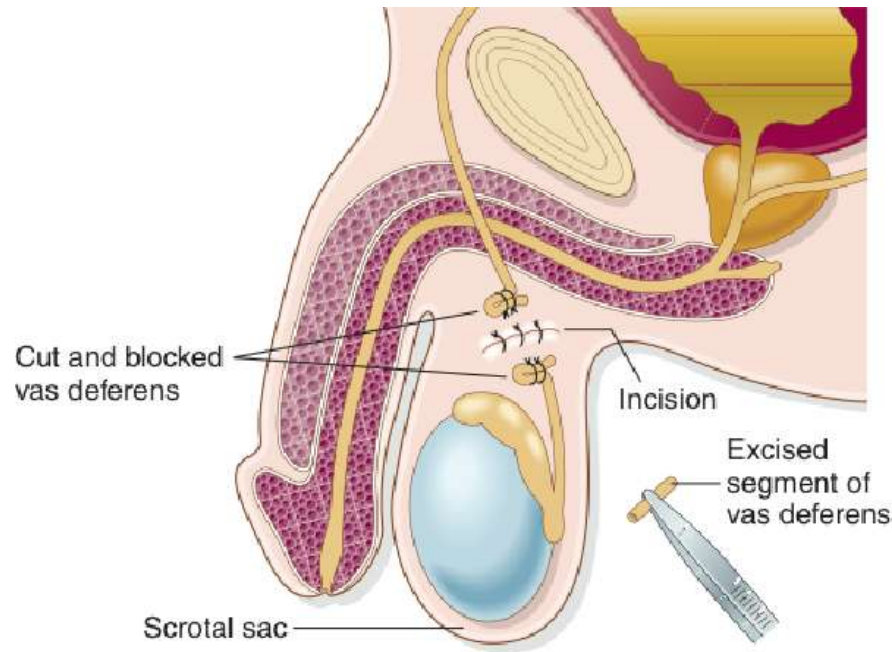


Laboratory Tests and Clinical Procedures

(Slide 2 of 2)

Clinical procedures

➤ Vasectomy



QUICK QUIZ (Slide 3 of 4)

3. Which term means removal of the prostate?

- A. Orchiectomy
- B. Prostatectomy
- C. Vasectomy
- D. Prostatitis

Abbreviations (Slide 1 of 2)

Abbreviation	Meaning
BPH	benign prostatic hyperplasia
DRE	digital rectal examination
ED	erectile dysfunction
GU	genitourinary
HPV	human papilloma virus
HSV	herpes simplex virus
PID	pelvic inflammatory disease
PIN	prostatic intraepithelial neoplasia
PSA	prostate-specific antigen

Abbreviations (Slide 2 of 2)

Abbreviation	Meaning
RPR	rapid plasma reagin
STI	sexually transmitted infections (diseases)
TRUS	transrectal ultrasound
TUIP	transurethral incision of the prostate
TUMT	transurethral microwave thermotherapy
TUNA	transurethral needle ablation
TURP	transurethral resection of the prostate

Review Sheet – Combining Forms

(Slide 1 of 4)

Combining Form
andr/o
balan/o
cry/o
crypt/o
epididym/o
gon/o
hydr/o
orch/o, orchi/o

Review Sheet – Combining Forms

(Slide 2 of 4)

Combining Form	Meaning
andr/o	male
balan/o	glans penis
cry/o	cold
crypt/o	hidden
epididym/o	epididymis
gon/o	seed
hydr/o	water, fluid
orch/o, orchi/o	testis, testicle

Review Sheet – Combining Forms

(Slide 3 of 4)

Combining Form
pen/o
prostat/o
semin/i
sperm/o
terat/o
test/o
varic/o
vas/o
zo/o

Review Sheet – Combining Forms

(Slide 4 of 4)

Combining Form	Meaning
pen/o	penis
prostat/o	prostate gland
semin/i	semen, seed
sperm/o	spermatozoa, semen
terat/o	monster
test/o	testis, testicle
varic/o	varicose veins
vas/o	vessel duct
zo/o	animal life

Review Sheet – Suffixes (Slide 1 of 2)

Suffix
-genesis
-one
-pexy
-stomy

Review Sheet – Suffixes (Slide 2 of 2)

Suffix	Meaning
-genesis	formation
-one	hormone
-pexy	fixation, put in place
-stomy	new opening

QUICK QUIZ (Slide 4 of 4)

4. Which term means “no sperm or semen are produced”?

- A. Aspermia
- B. Oligospermia
- C. Anorchism
- D. Spermolytic